

Problem

Design a problem to help other students better understand ω_0 , Q , and B at resonance in series RLC circuits.

Step-by-step solution

Step 1 of 1

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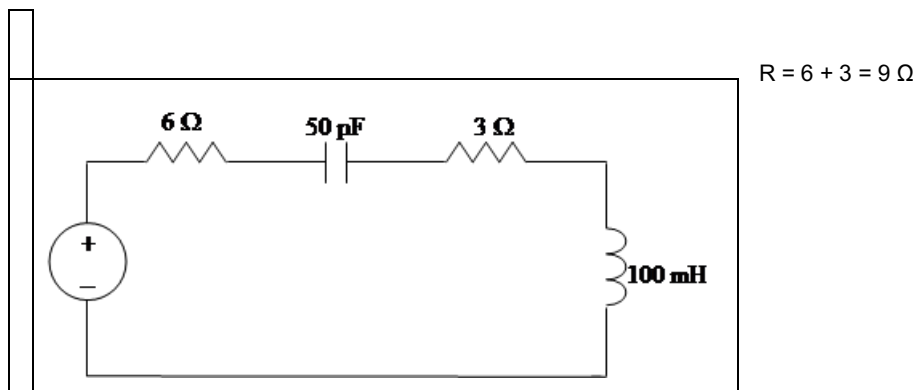
Although there are many ways to solve this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

A coil with resistance $3\ \Omega$ and inductance 100 mH is connected in series with a capacitor of 50 pF , a resistor of $6\ \Omega$, and a signal generator that gives 110V-rms at all frequencies. Calculate ω_0 , Q , and B at resonance of the resultant series RLC circuit.

Solution

Consider the circuit as shown below. This is a series RLC resonant circuit.



$$\omega_o = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{100 \times 10^{-3} \times 50 \times 10^{-12}}} = \underline{447.21\text{ krad/s}}$$

$$Q = \frac{\omega_o L}{R} = \frac{447.21 \times 10^3 \times 100 \times 10^{-3}}{9} = \underline{4969}$$

$$B = \frac{\omega_o}{Q} = \frac{447.21 \times 10^3}{4969} = \underline{90\text{ rad/s}}$$